STATS C161 / C261

UCLA · MIDTERM PROJECT · PROFESSOR CHENG

DATE & TIME
Oct 29, 2026
15 Min Presentation

Can AI Beat the Crowd?

“Is a prediction-market price a good probability, and can a model do better?”

Testing Market Efficiency, Favorite-Longshot Bias, and Wallet Micro-Behaviors on Polymarket

DATASET
7,479
resolved markets on Polymarket

WALLETS
55,952
Tracked Profiles

VOLUME
4.1M
trades used to read prices

By Leo Wong · Jiahong Chen · Aaron Wen
Lit · Market · Wallet

[Excused Note: Group consists of 3 undergraduate, 1 graduate dropped]

The Story Arc: From Market Prices to Trader Intelligence

ACT I

Market Efficiency

Prices on Polymarket are exceptionally well-calibrated out of sample. Predictive models cannot beat raw prices.

ACT II

Wallet Heterogeneity

If the market is efficient overall, who buys low-probability longshots? Gradient boosting isolates key wallet traits.

BRIDGING TO FINAL

Informed Detection

Combining market anomalies with wallet profiling to detect informed/insider trading in Politics & Tech.

BACKGROUND & PRIOR RESEARCH · LEO

SPEAKER: LEO (1:00)

Academic Foundations: What past research says

Thaler and Ziemba (1988)

Journal of Economic Perspectives 2(2)

"Surveys racetrack and lottery betting markets and the pricing anomalies found there."

Snowberg and Wolfers (2010)

Journal of Political Economy 118(4)

"Favorite-longshot bias: longshots are overbet and favorites underbet. Misperceived probabilities fit best."

Wolfers and Zitzewitz (2004)

Journal of Economic Perspectives 18(2)

"Prediction-market forecasts are typically fairly accurate and beat many simple benchmarks."

Page and Clemen (2013)

Economic Journal 123(568)

"Prices are reasonably calibrated close to expiry, but biased for events farther away."

Our test: do these findings hold on Polymarket in 2026, from 7 days down to 6 hours before close?

Leo · Sources verified: JEP 2(2):161-174; JPE 118(4):723-746; JEP 18(2):107-126; Economic Journal 123(568):491-513

A price is a probability

$$E[\text{profit}] = p - c$$

p is your belief, c is the share price. Buy while p > c, sell while p < c, so the price settles near the crowd's belief.

1 · Is the price calibrated?

Does 30% happen 30% of the time?

2 · Can a model beat it?

Using only information known at the time.

Classic finding from betting markets: the favorite-longshot bias, where long shots are overpriced. We test whether it shows up here.

Data Methodology & Pipeline

Resolved markets

7,479 Yes/No markets, Politics, Technology, Entertainment and Economy, closing Jun to Oct 2026. Categories from Polymarket event tags.

Price before close

4.1M trades. We read the last trade before 6 hours, 24 hours, 3 days and 7 days to close.

Grade against outcome

Brier score and calibration curves, with confidence intervals that resample whole events.

Order book recorded only about 22% of these markets. Where both exist, trade price and book midpoint agree: correlation 0.92, median gap 0.9 cents.

7,479	4.1M	4	0.92
Resolved Markets	Trade Records	Horizons (6h, 24h, 3d, 7d)	Order-Book Correlation

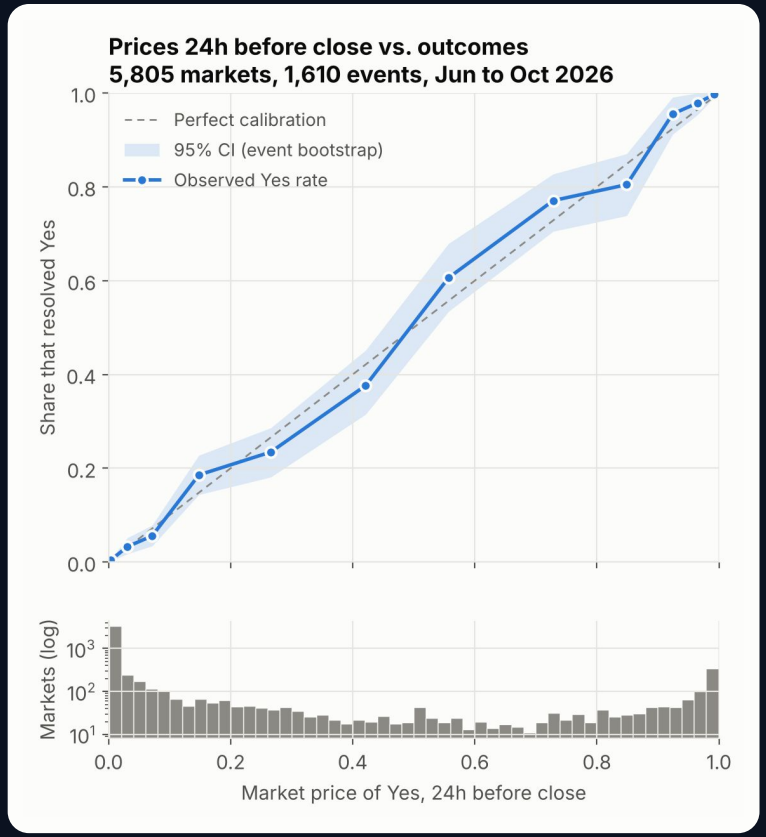
Rigorous Validation Framework

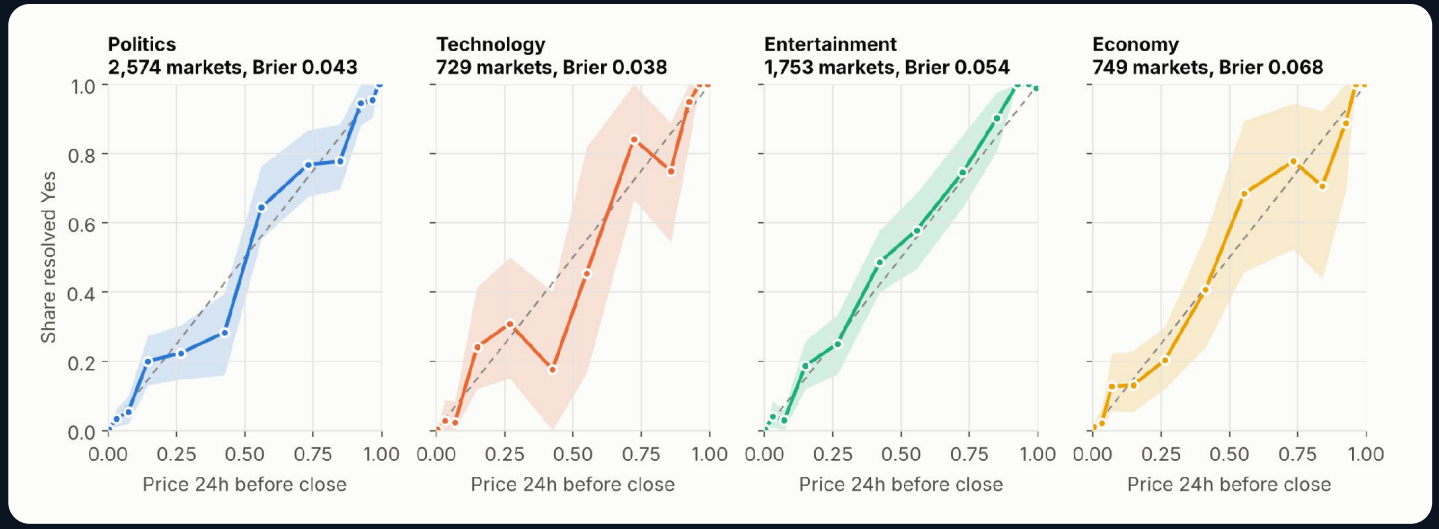
Brier score metrics evaluated using event-level block bootstrap confidence intervals.

`Median Gap: 0.9¢`

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Calibration: the crowd is well calibrated





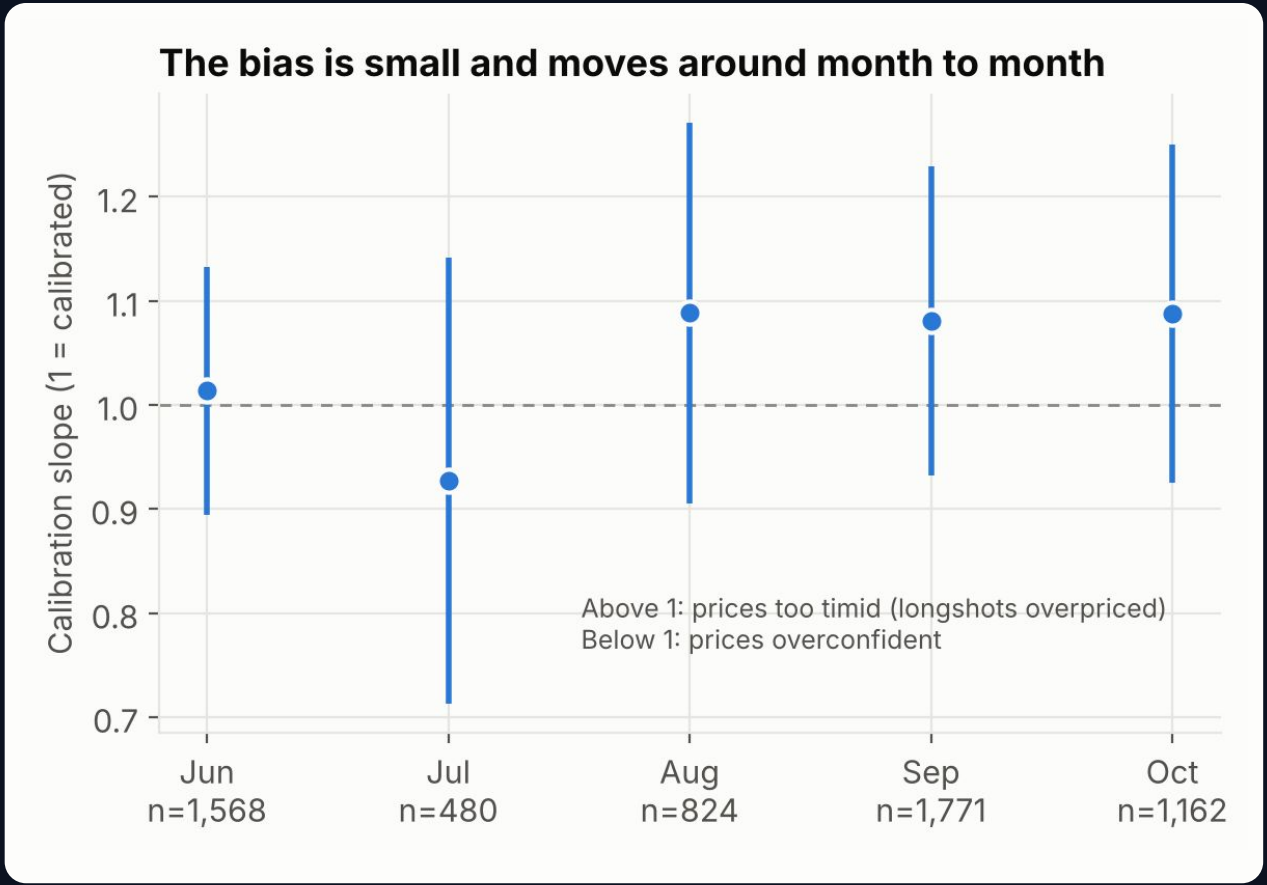
CORE FINDING

Near-Perfect Alignment Across Topics

When the market says 30%, it happens about 30% of the time. The same holds in each category.

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Favorite-longshot bias



1.08 Raw Calibration Slope

calibration slope at 6h, all prices (95% CI 1.01 to 1.16)

1.00 Trimmed Slope (<2¢ / >98¢)

slope when prices under 2¢ and over 98¢ are dropped

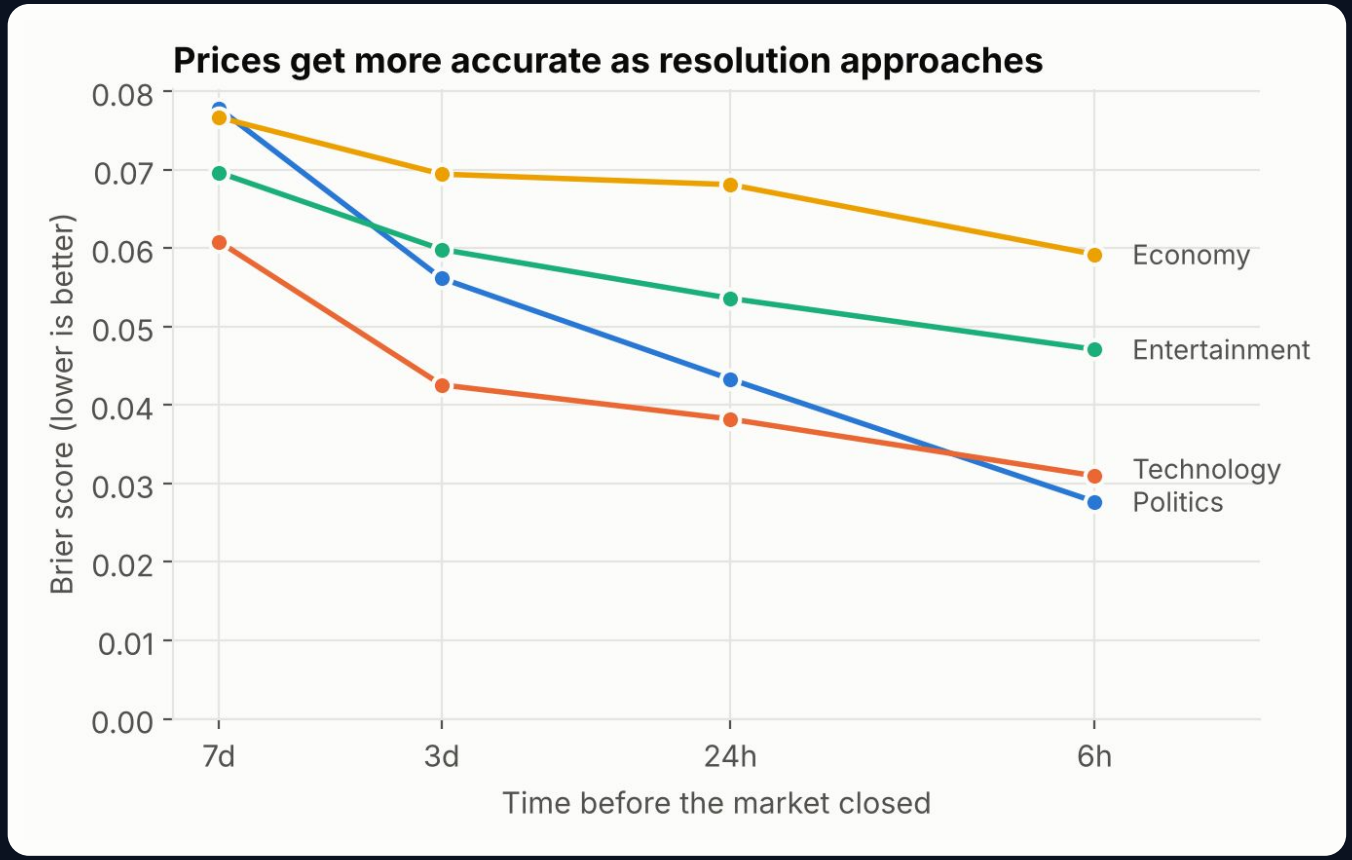
0.93 to 1.09

Monthly Slope Range

monthly slope at 24h; every interval crosses 1

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Accuracy over time and by category



0.074 → 0.038

Brier score, all markets, 7 days to 6 hours before close

Politics improves the most. From 3 days out to the close, Economy is the hardest category to call.

Brier = average squared gap between price and outcome. Lower is better.

Politics: Fastest Convergence

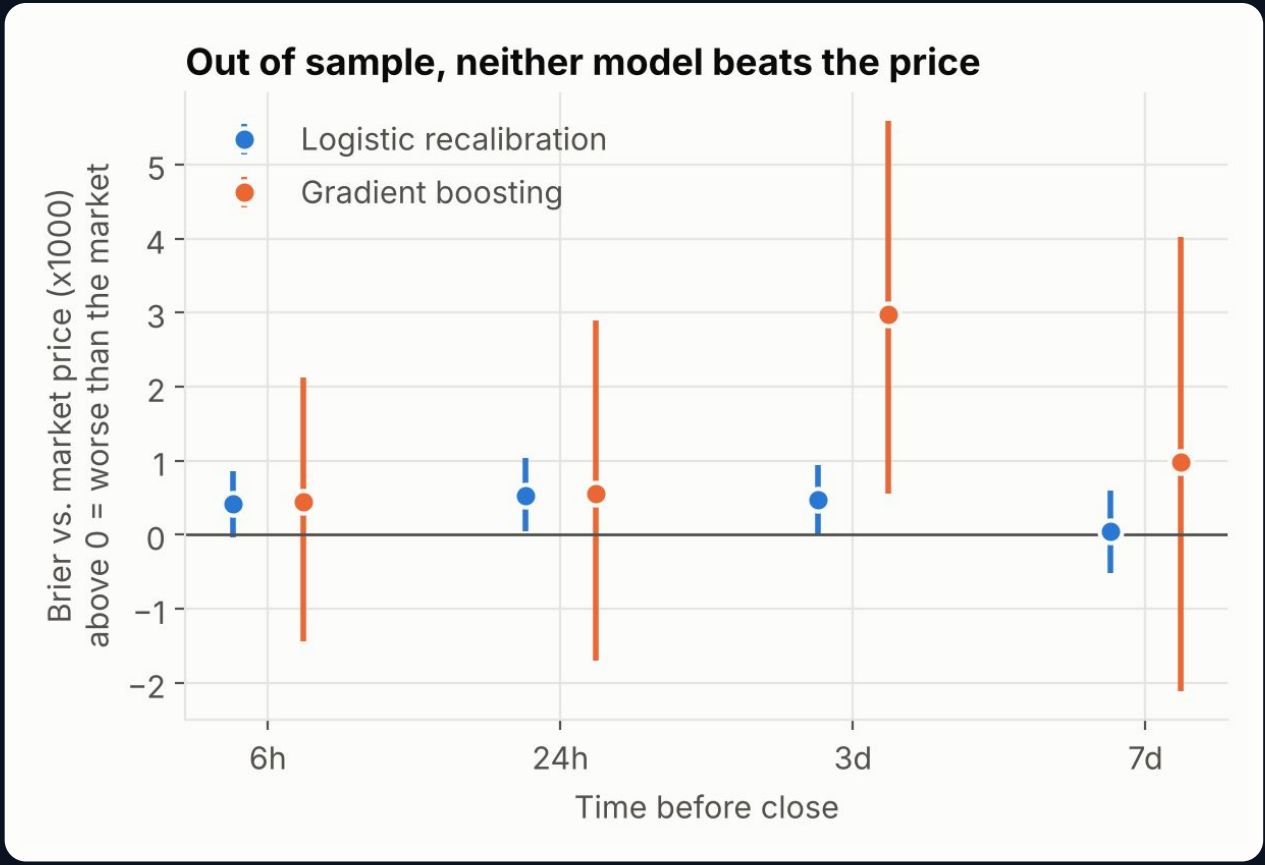
Public polling data accelerates price discovery ahead of deadlines.

Economy: Highest Residual Error

Macro metrics maintain high variance right up to official releases.

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Can a model beat the price?



Test Brier at 24h

Above zero in the chart = worse than the market.

Raw Market Price Baseline Benchmark	0.0513
Logistic Recalibration +18 Features	0.0518
Gradient Boosting (LightGBM) Full Book & Volume Features	0.0518

Core Takeaway: No Overperformance

Trained on pre-Sept data and evaluated out-of-sample on 860 Sept–Oct events, machine learning models failed to outperform simple market prices.

```
Tree Weight on Price Feature: >95%
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Takeaways and limits

1
Prices are well calibrated in all four categories.

2
The favorite-longshot bias is faint and unstable.

3
No model we built beats the price.

Limit: some markets close early because the event happened, so "24h before close" differs for Yes and No markets.

Limit: one test period, September to October 2026.

Conceptual Pivot to Act II

"Because prices are aggregate-efficient and un-scalable by standard ML models, **an informed trader is not simply someone who wins a bet**, but someone who systematically beats the prevailing market price."

`NEXT STEP:` Analyze wallet-level heterogeneity to isolate informed trading patterns.

Question and data

How well do six wallet traits predict a wallet's average purchase price?

Does gradient boosting beat linear regression?

55,952 wallets, one row each. Target: volume-weighted price paid for shares bought.



Reminder: "An informed trader is one who beats the price, not one who wins a lot."

Approach

Split

44,761 training wallets and 11,191 test wallets. One fixed random split; every method sees the same test wallets.

Models

Mean baseline, additive linear regression, and histogram gradient boosting for nonlinear patterns and interactions.

Inputs

Trade count, market count, activity span, concentration, total spending, largest purchase. Skewed inputs use log(1 + x).

Five-fold validation happens inside training only. Boosting is chosen from a six-setting grid by validation error; the test set stays untouched.

Data Partitioning

44,761 training wallets and 11,191 held-out testing wallets.

Models Evaluated

Mean Baseline vs. Additive Linear Regression vs. HistGradientBoosting.

Input Feature Matrix (Log-Transformed)

- Trade Count
- Market Count
- Activity Span
- Concentration
- Total Outlay
- Max Single Purchase

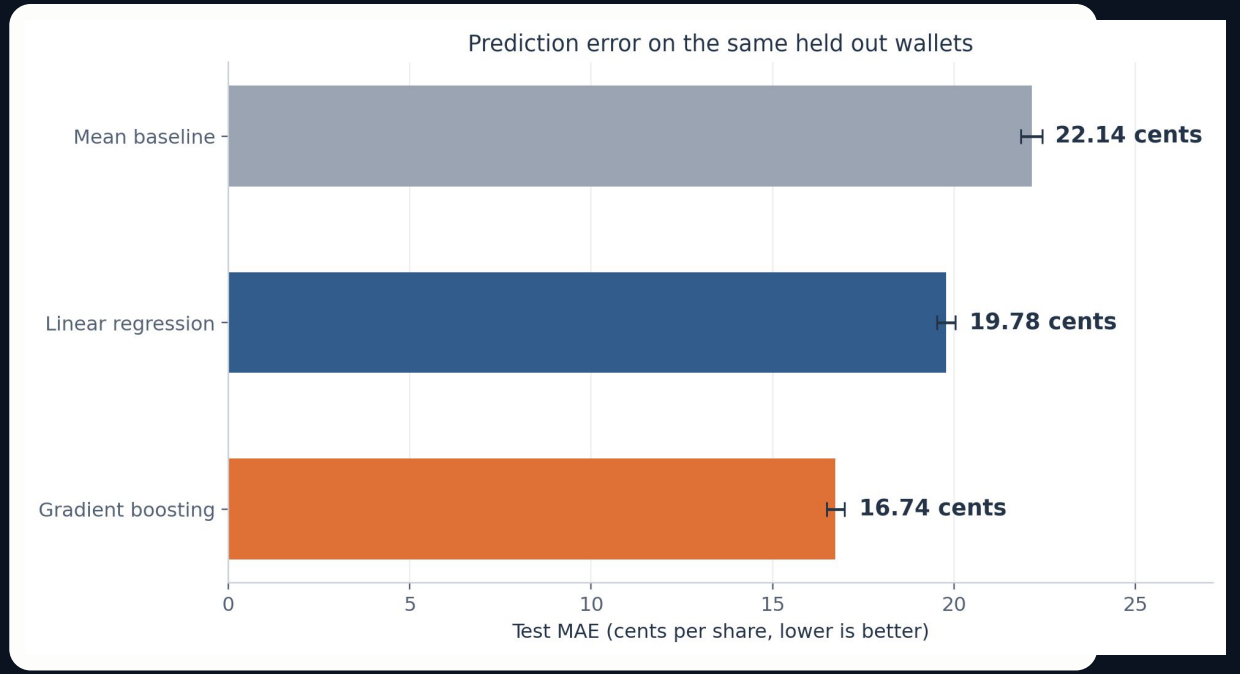
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Results

22.14¢
mean baseline, test MAE

19.78¢
linear regression

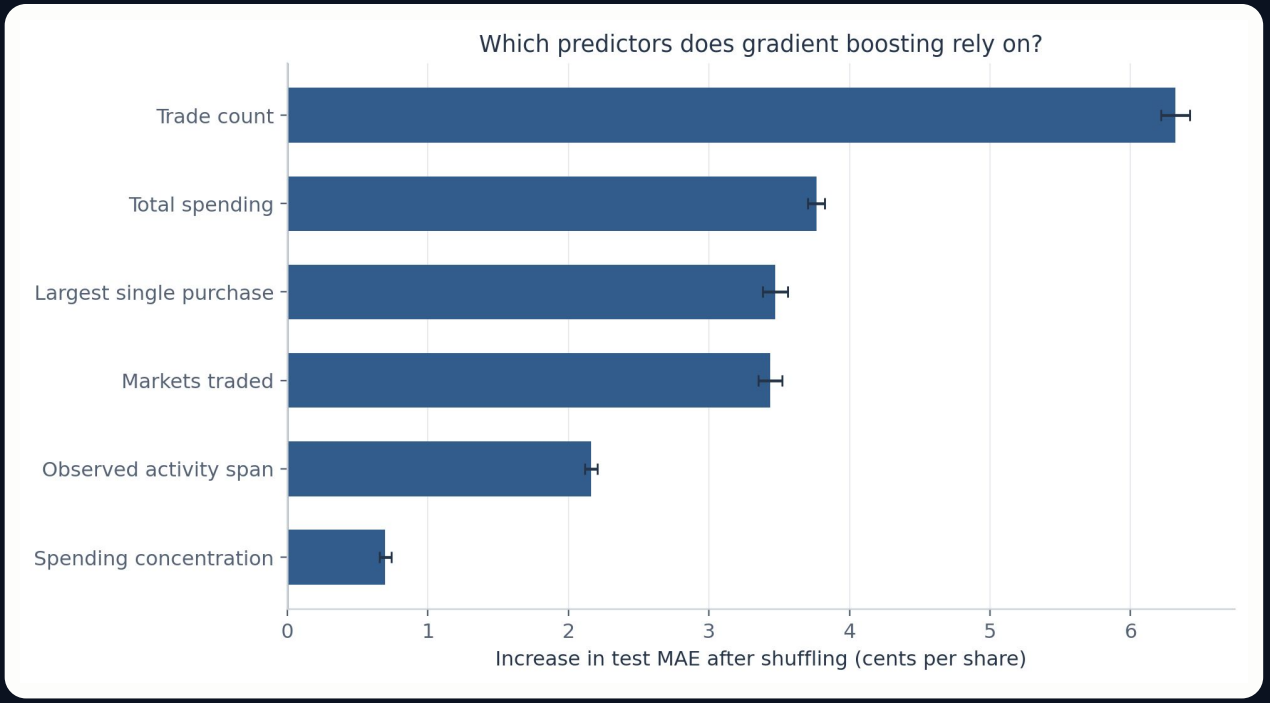
16.74¢
gradient boosting



Boosting beats linear by 3.04¢ (95% interval 2.86 to 3.23¢), a 15.4% improvement. R^2: 0.368 vs 0.197.

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Interpretation



Trade count matters most: shuffling it raises test error by about 6.32¢. `+6.32¢ MAE Impact`

Removing spending and largest purchase raises boosting error from 16.74¢ to 19.97¢.

This is descriptive: not profit, trading skill or cause.

Total Spending & Max Purchase `High Importance`

Secondary indicators driving trading profile separation.

Scope Limitation Note

This model captures **behavioral archetypes** (longshot bias vs. favorite buyers) rather than serving as a direct causal indicator of net profitability or trading skill.

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Thank You!

Questions & Comments?

We welcome feedback on our calibration methodology and wallet profiling pipeline.

Q&A Session (~3 mins)

Leo Wong
Literature & Deck

Aaron Wen
Market Calibration

Jiahong Chen
Wallet Analysis

BACKUP · FOR Q&A

Details if asked

Train and test

Train: 841 events, 3,215 markets, all closed before Sep 1. Test: 860 events, 3,511 markets. An event that straddles Sep 1 goes wholly to test.

Mid-range prices

Only prices from 5¢ to 95¢: test Brier at 24h is 0.148 for the market and 0.150 for both models.

Slopes by category

At 24h: Politics 1.03, Technology 1.11, Entertainment 1.12, Economy 0.92. Only Entertainment's interval sits just above 1 (1.00 to 1.29).